Update on tracking efficiencies

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Introduction

Strategy for tracking efficiency determination:

- Tag-and-probe method with $J/\psi \to \mu^+\mu^-$
- Combine efficiencies of two methods:
  - VeloMuon: SciFi efficiency
  - Downstream: Velo efficiency
  - Combined: Velo $\times$ SciFi efficiencies
- Cross check with long track efficiency with the MuonUT method
  - Main systematic

Tag-and-probe method

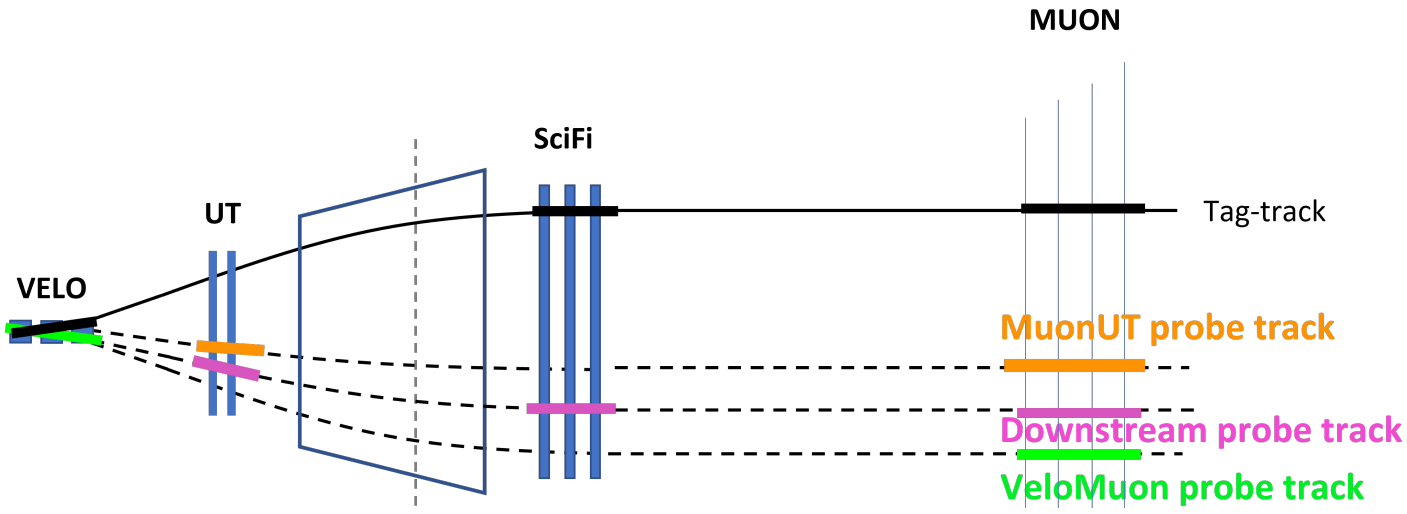

Figure by Rowina

- Fully reconstruct one muon from $J/\psi \to \mu^+\mu^-$
- Partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Overlap criteria in matching

Matching is done by comparing the hits of the probe track with a long track in each subdetector

Method	Velo	UT	SciFi	Muon
VeloMuon	$> 40\%$	–	–	$> 40\%$
Downstream	–	–	$> 40\%$	$> 40\%$
MuonUT	–	0 or $> 40\%$	–	$> 40\%$

Note:

- Matching was originally performed in a trigger line
- We now have functors for direct access to overlap fractions
- Requires rerunning reconstruction in 2024, as long-track information was not persisted

Overview of previous work

What I'm showing today is based on Rowina's thesis work

- Previous presentation here
- Rowina's thesis here

Today's results are from TrackCalib2, which was maintained by Maurice

- GitLab repository here
- Fitter independent from Rowina's code, but results are consistent
- Tuples are currently fetched directly from an AP

Tracking efficiency plan

We are providing analysts the Data/MC long track efficiency ratio for MC reweighting/correction:

- Long track efficiency from product of Velo and SciFi efficiencies
- 2D p–η binning MC reweighting in physics analysis
- Statistical uncertainties from limited data/MC sample sizes
- Dominant systematic uncertainty from MuonUT-Combined difference

Overview of current progress

What I will talk about today on tracking efficiencies:

1. Block 1a vs 1b studies
2. Results for tracking efficiencies
   - 2024 blocks 2, 1a, 5–8
   - 2024 *pp* ref run
   - 2025 pre-TS
3. Guide to TrackCalib2

Overview of current progress

What I will talk about today on tracking efficiencies:

1. **Block 1a vs 1b studies**
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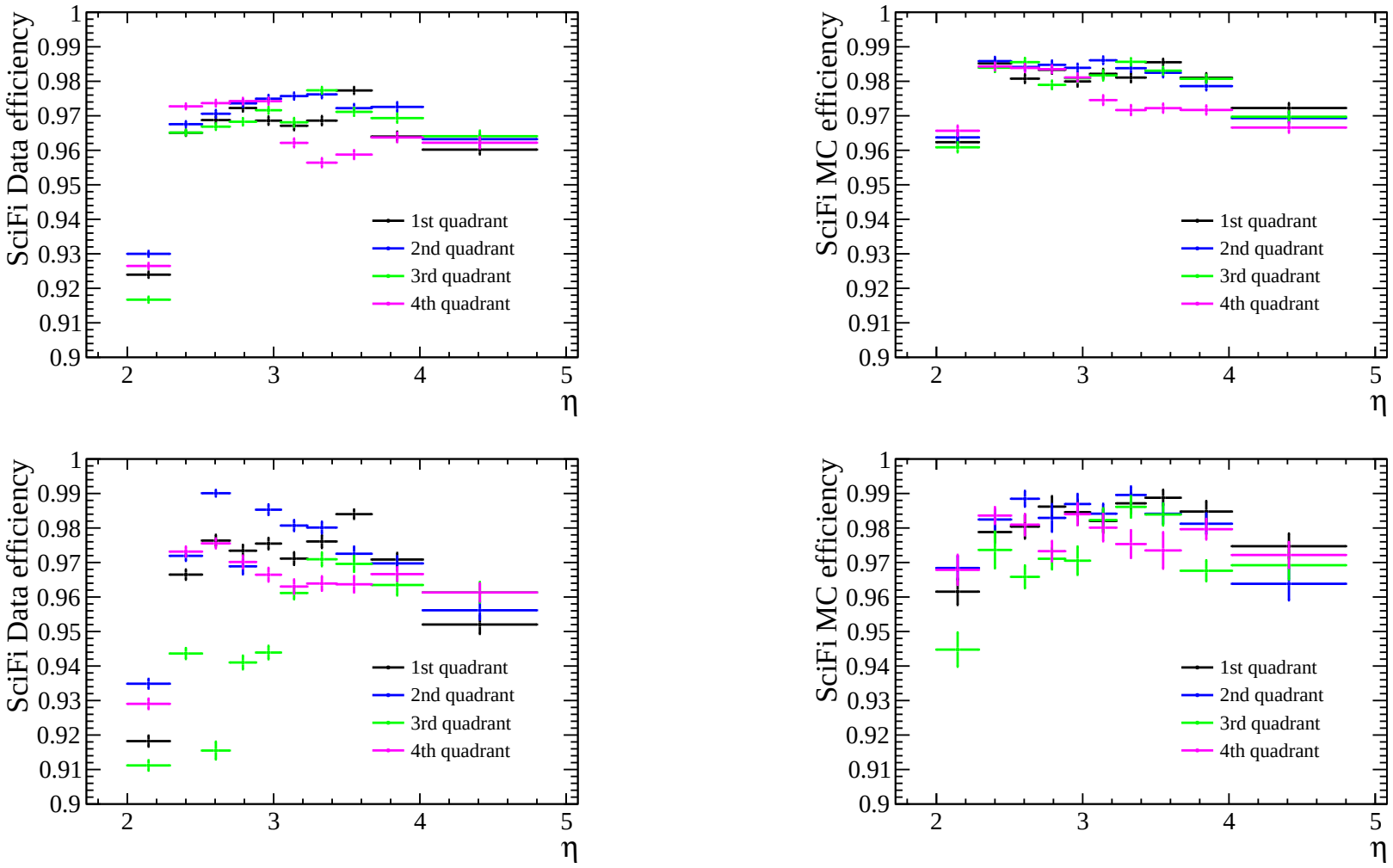
Block 1a vs 1b

Block 1a (fills 9982–10012, top) vs 1b (fills 10017–10056, bottom)



SciFi T2L0Q0M1H0 FE was excluded on 17th August

Block 1a vs 1b

MC doesn't seem to describe the drop in tracking efficiency well...

- We will check with Simulation people
- Note: The "hole" in the SciFi is only visible when looking at tracks in the 3rd quadrant, the integrated numbers show negligible differences between block 1a and block 1b
- We propose that all tracking efficiencies are only provided for block 1a
- Analyses that are sensitive to this missing SciFi FE should exclude block 1b, other analyses can use results for block 1a as a proxy for block 1b

Overview of current progress

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Remarks on current results

Remark on resprucing of 24c4:

- 40% prescale present in sprucing
- Very limited statistics, especially in MuonUT
- Turcal resprucing campaign (24r1) started 3 weeks ago...
- ... but a bug in the merging step delayed the campaign until last week
- Results shown for 24c4, to be updated with 24r1 in a few weeks

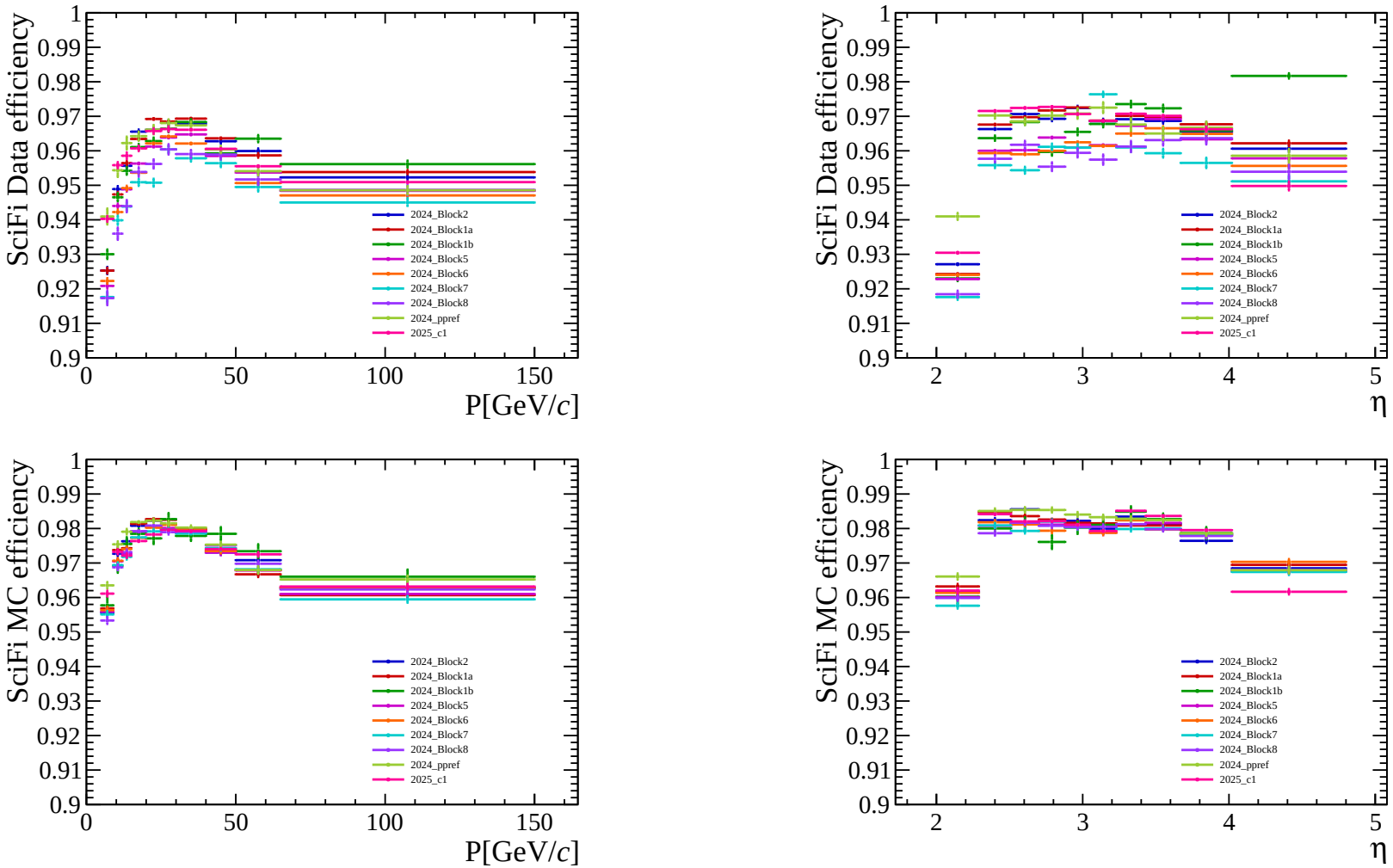
Remarks on current results

Remark on phase-space differences between 2024 and 2025:

- Probe muon p_T cut was at 500 MeV in 2024
- Loosened to 100 MeV in 2025
- Therefore cannot compare 2024 and 2025 directly

Efficiency comparisons

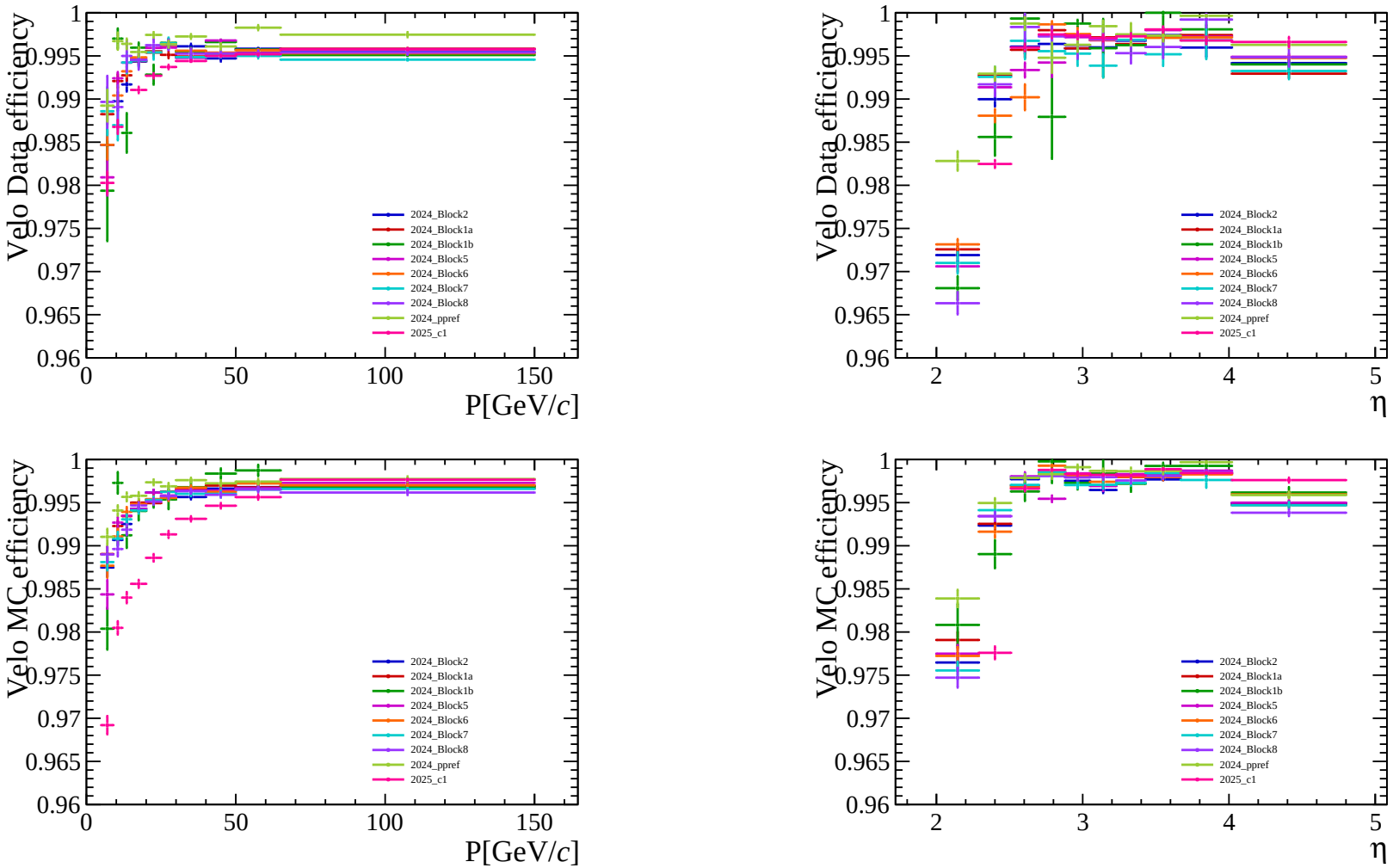
Comparison of tracking efficiencies in all data blocks



SciFi efficiencies

Efficiency comparisons

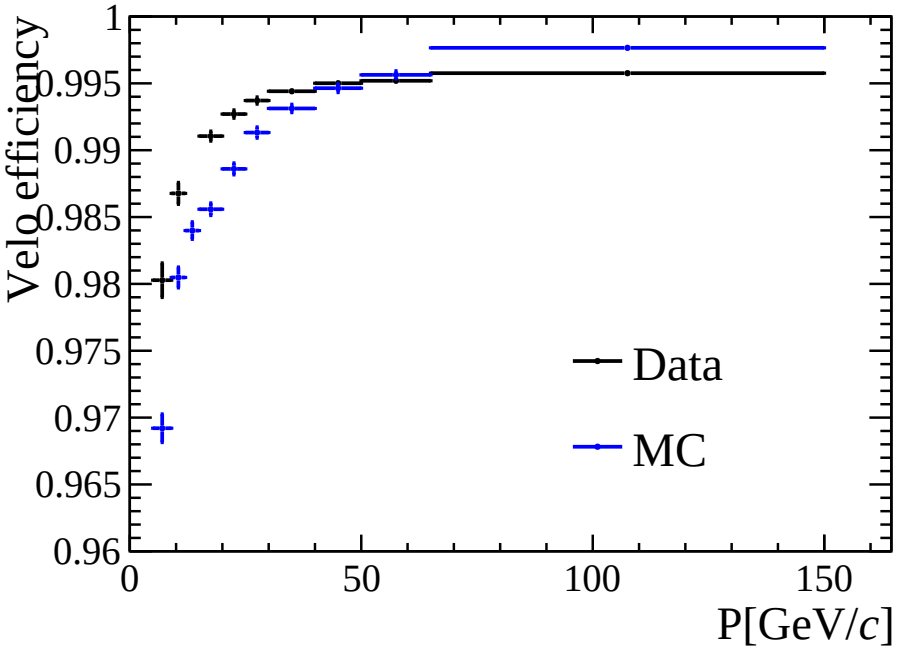
Comparison of tracking efficiencies in all data blocks

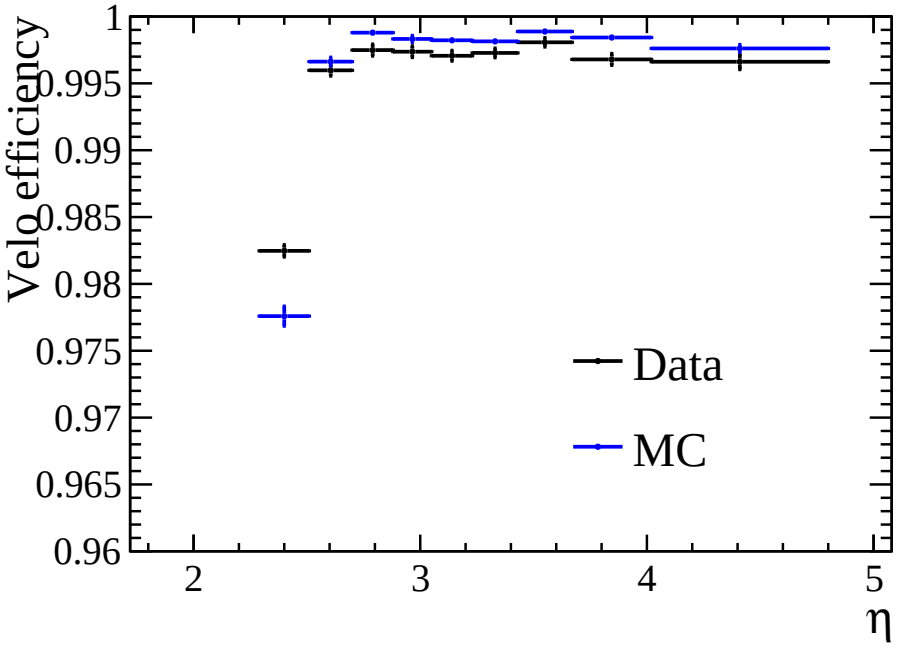


Velo efficiencies

Efficiency comparisons

2025 Data-MC discrepancy

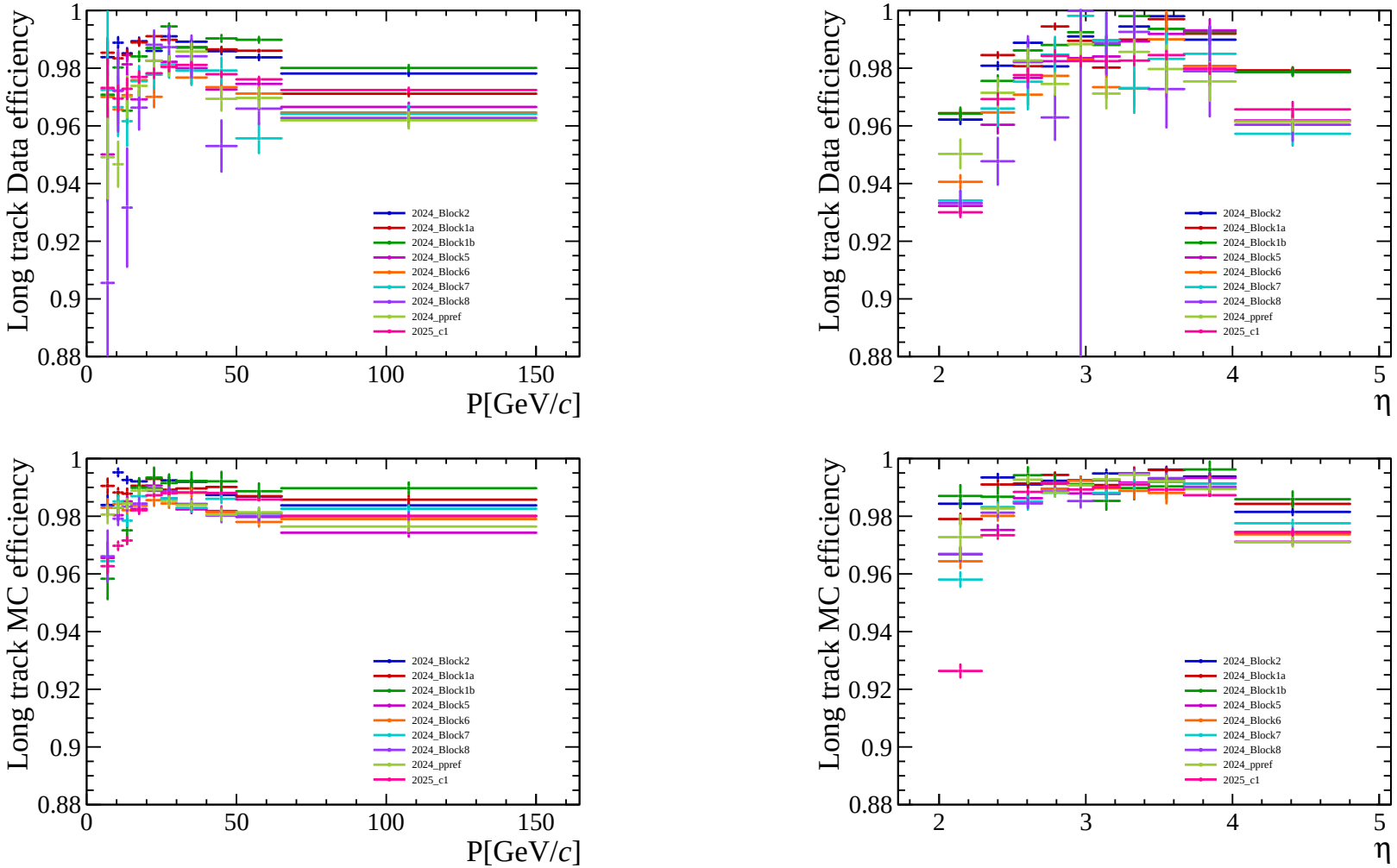




Need to understand why MC underestimates the Velo efficiency in 2025!

Efficiency comparisons

Comparison of tracking efficiencies in all data blocks



Long track efficiencies

General observations

General observations on tracking efficiencies:

1. All data sets show consistent trends
2. Performance seems to (slowly) drop over time

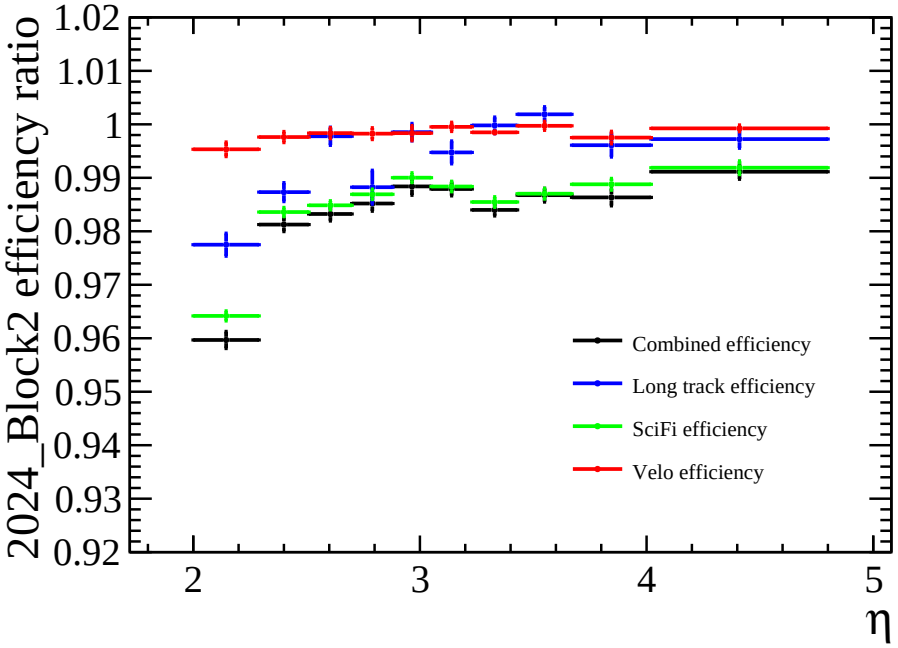
This is very encouraging! Several issues resolved, including:

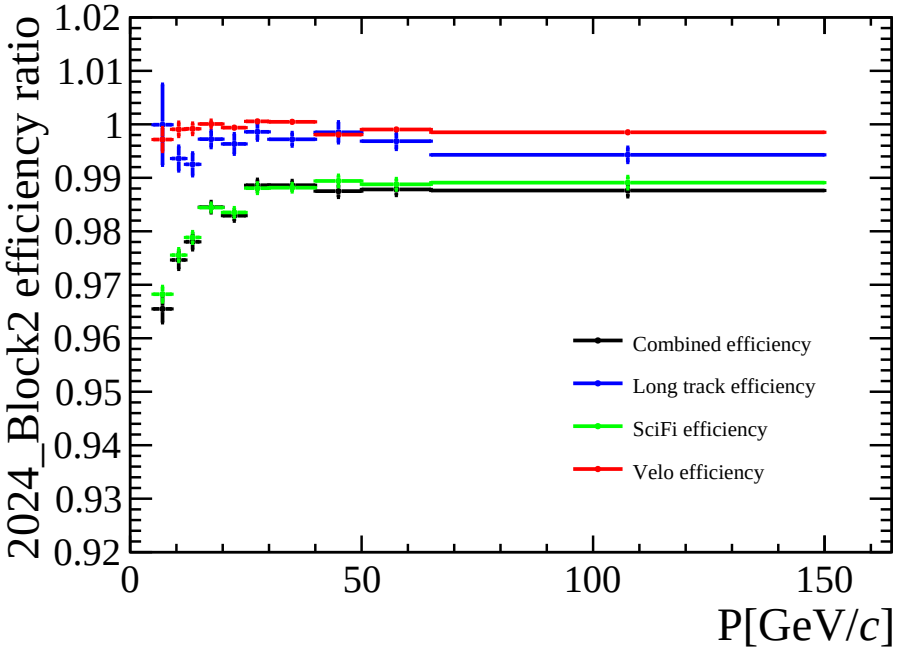
1. Unusually high efficiencies in blocks 7/8
2. Missing binds when rerunning reconstruction
3. Large charge asymmetries in MuonUT samples

However, the low Velo efficiency in MC needs to be understood, so we will wait with the release of these numbers

Tracking efficiency data/MC ratio

Data/MC ratio

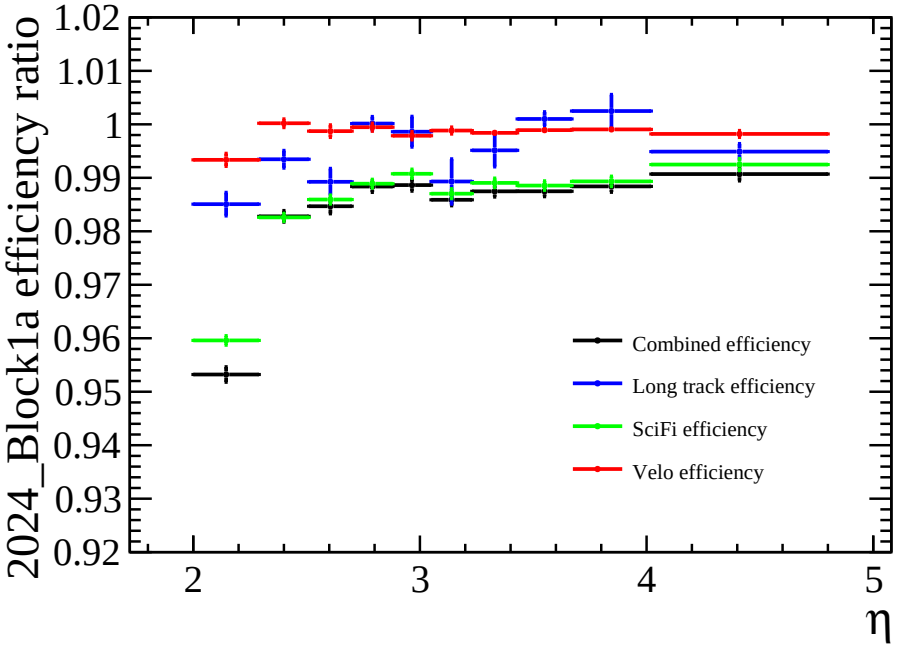


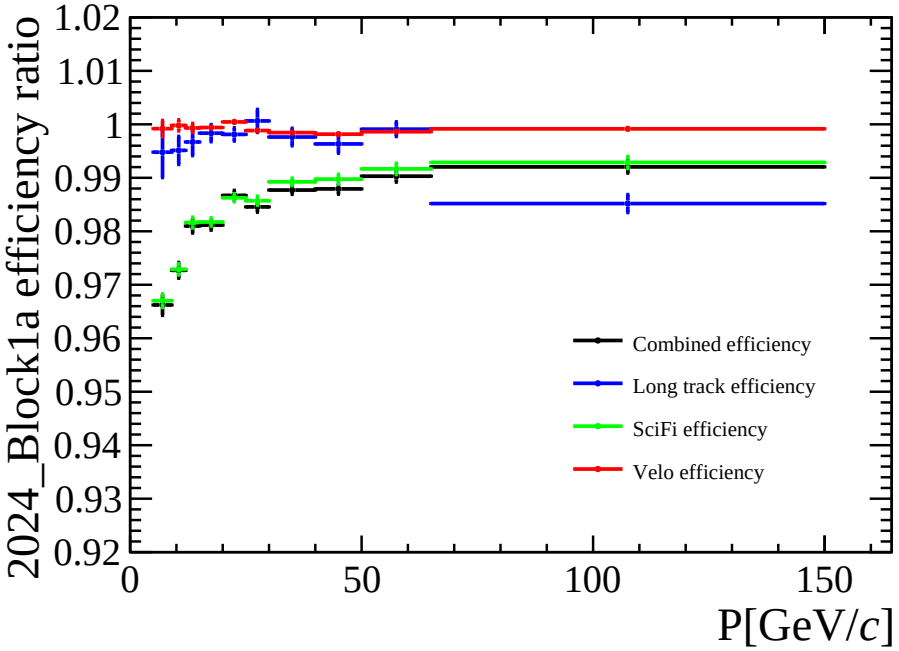


Data/MC ratio for Combined/MuonUT track efficiencies block 2

Tracking efficiency data/MC ratio

Data/MC ratio

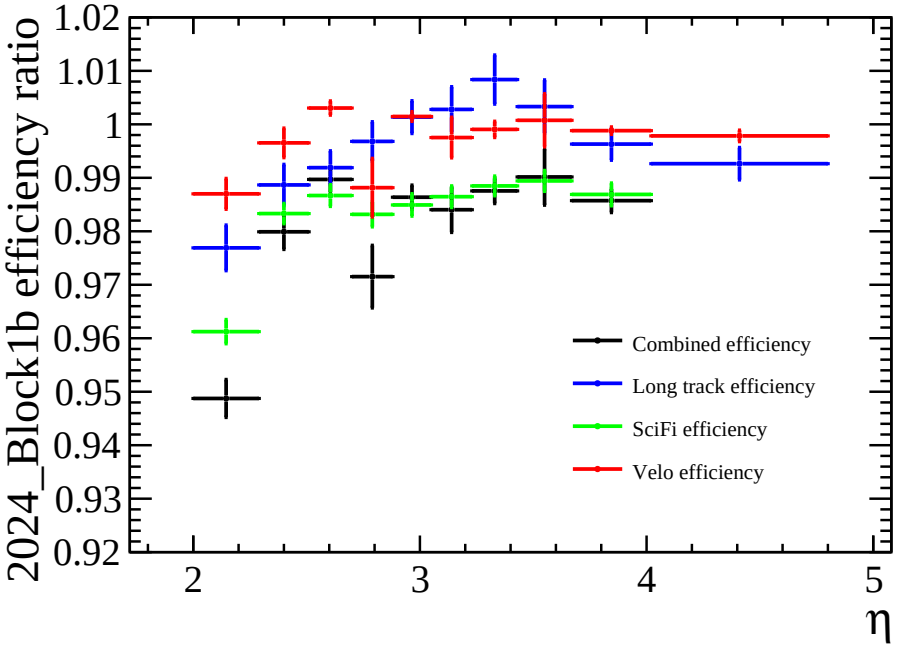


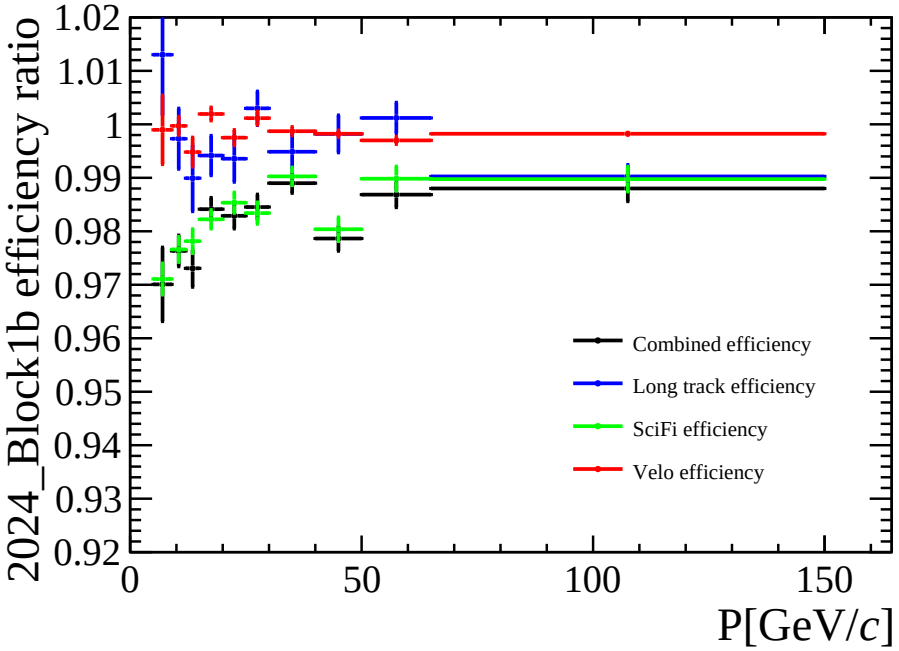


Data/MC ratio for Combined/MuonUT track efficiencies block 1a

Tracking efficiency data/MC ratio

Data/MC ratio

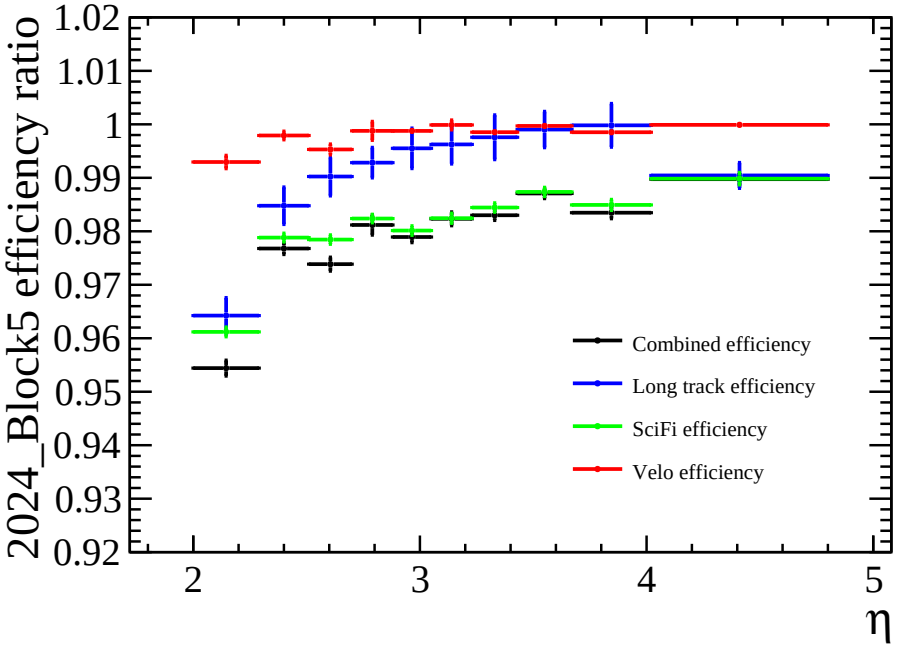


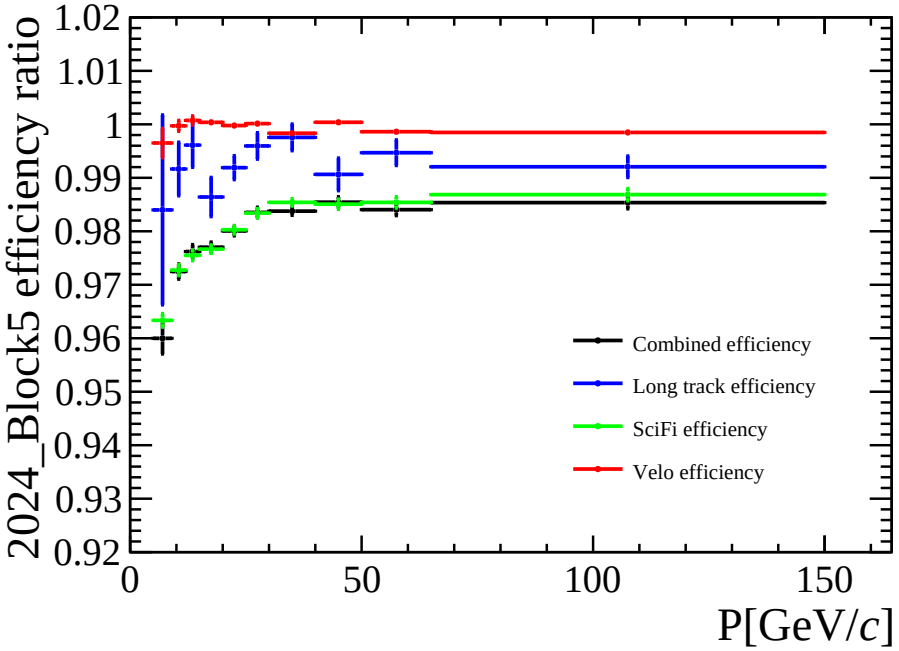


Data/MC ratio for Combined/MuonUT track efficiencies block 1b

Tracking efficiency data/MC ratio

Data/MC ratio

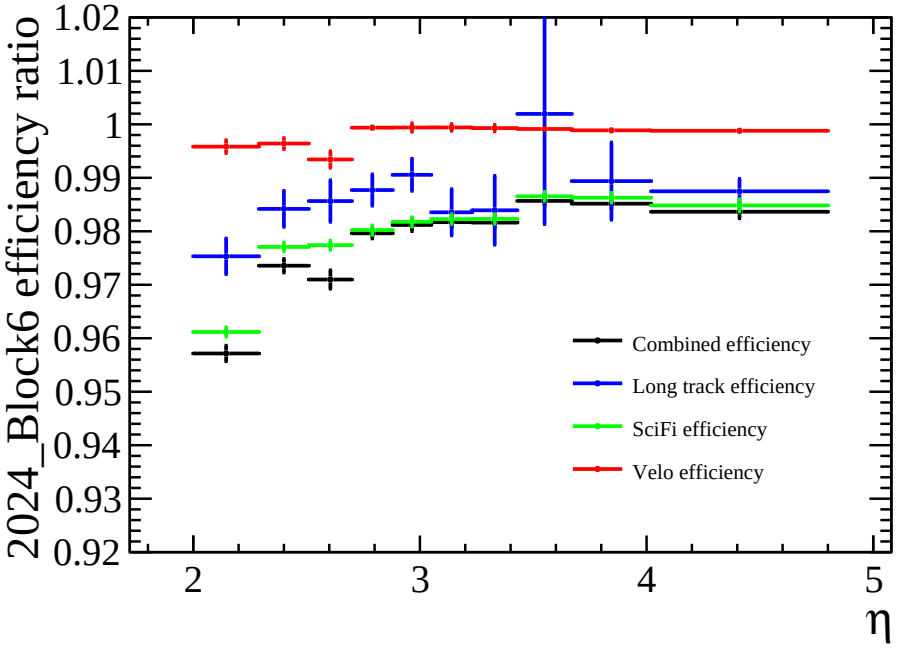


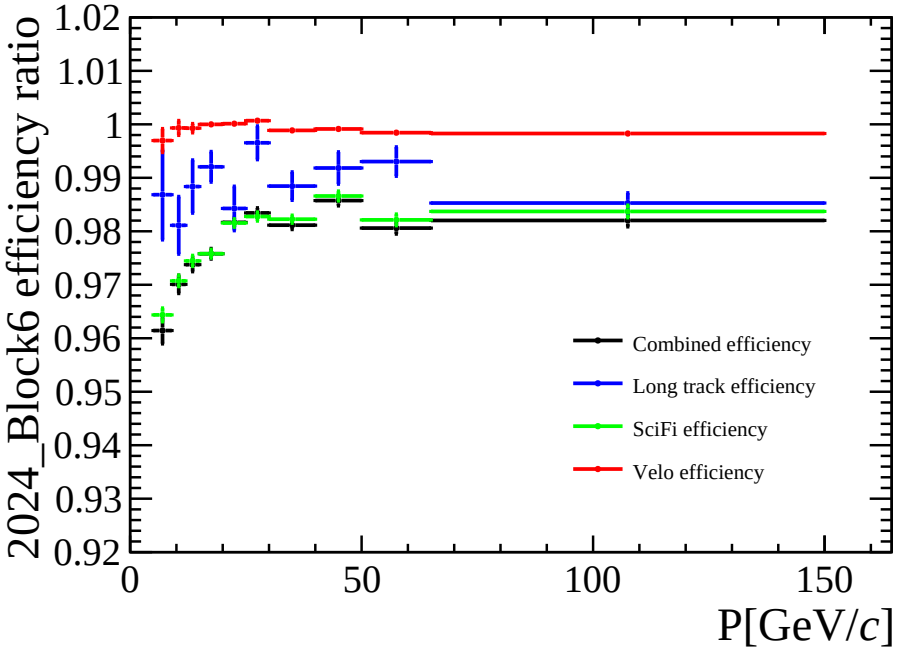


Data/MC ratio for Combined/MuonUT track efficiencies block 5

Tracking efficiency data/MC ratio

Data/MC ratio

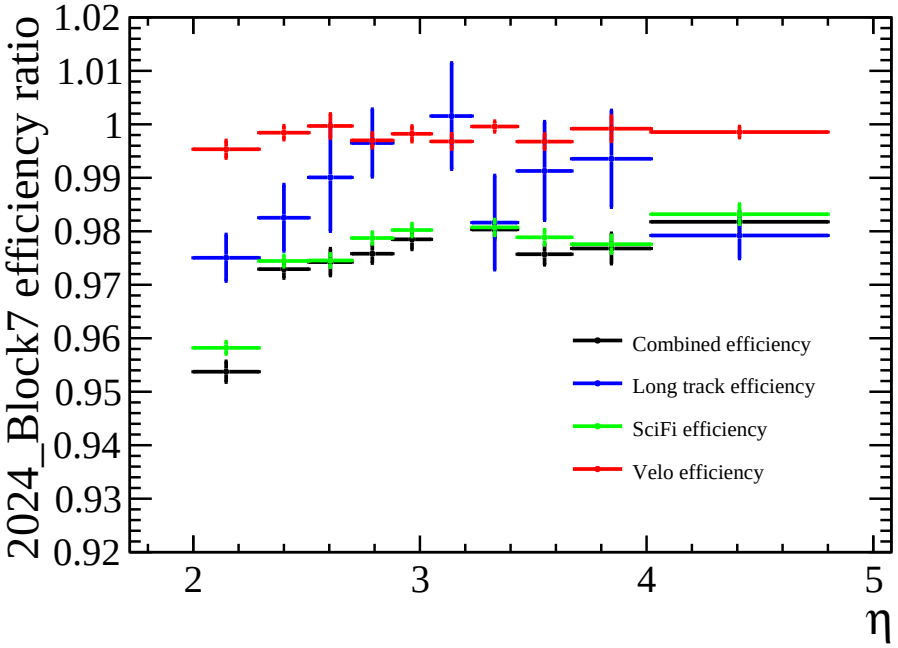


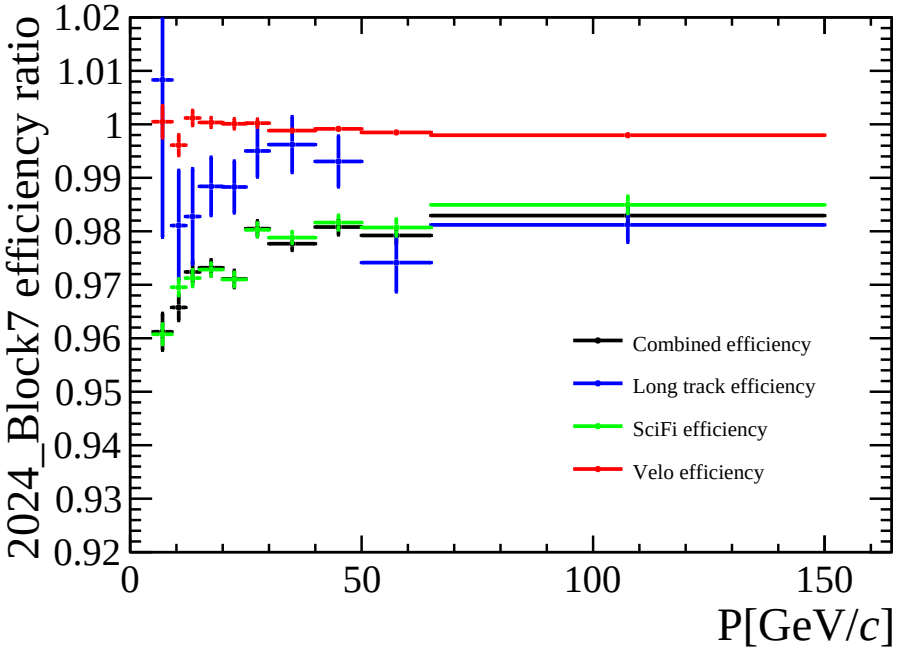


Data/MC ratio for Combined/MuonUT track efficiencies block 6

Tracking efficiency data/MC ratio

Data/MC ratio

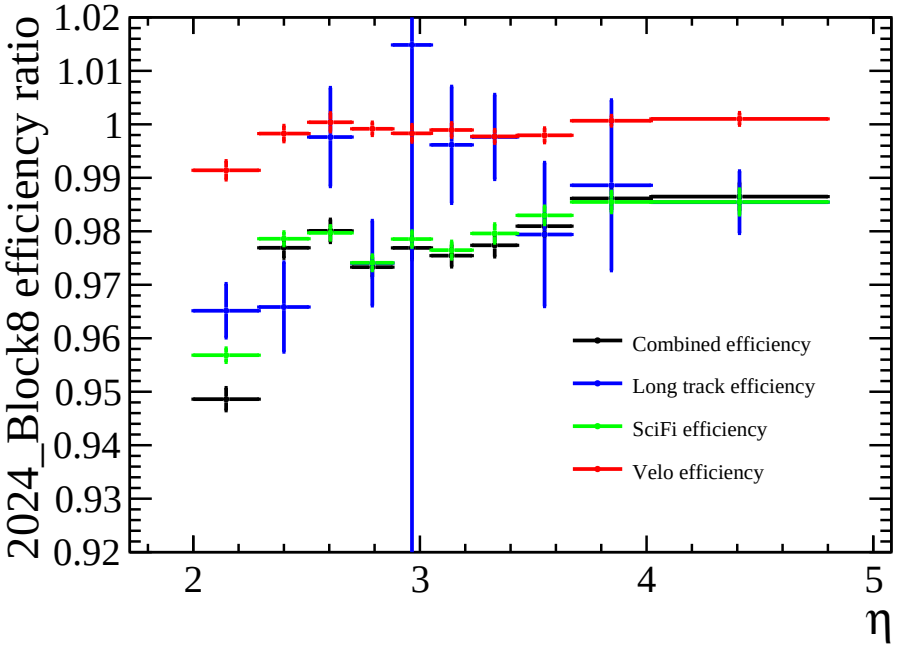


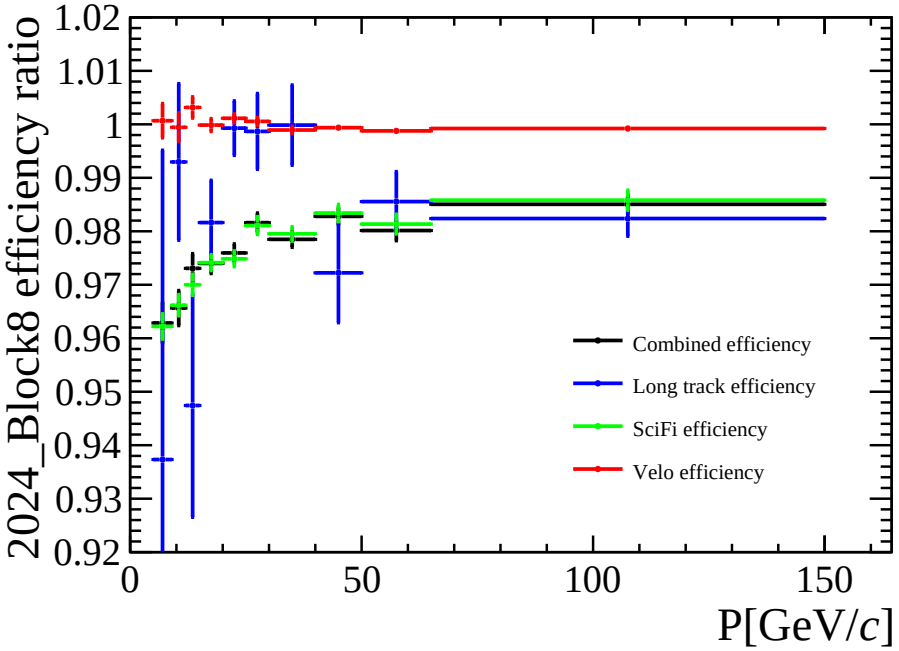


Data/MC ratio for Combined/MuonUT track efficiencies block 7

Tracking efficiency data/MC ratio

Data/MC ratio

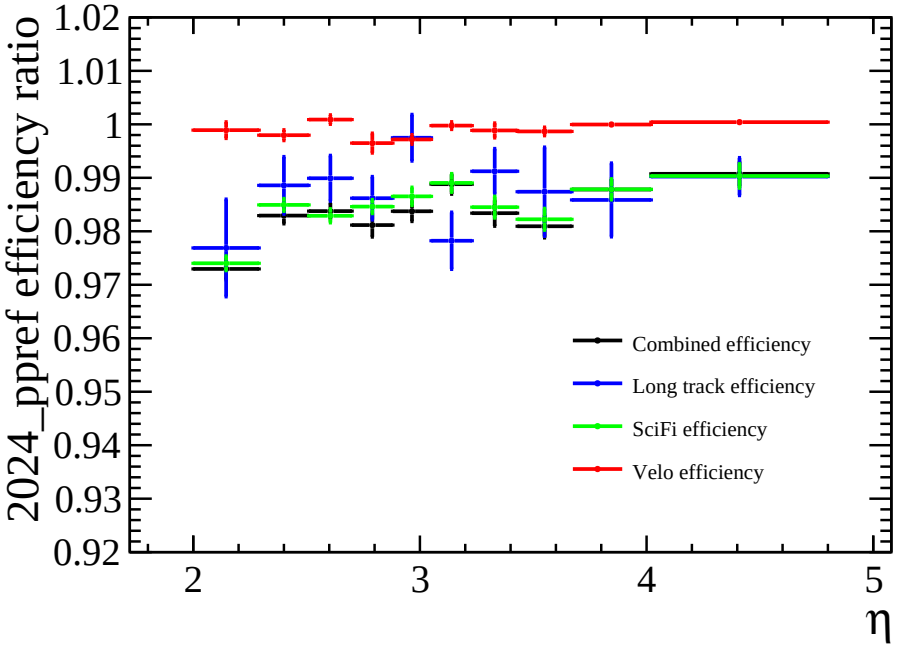


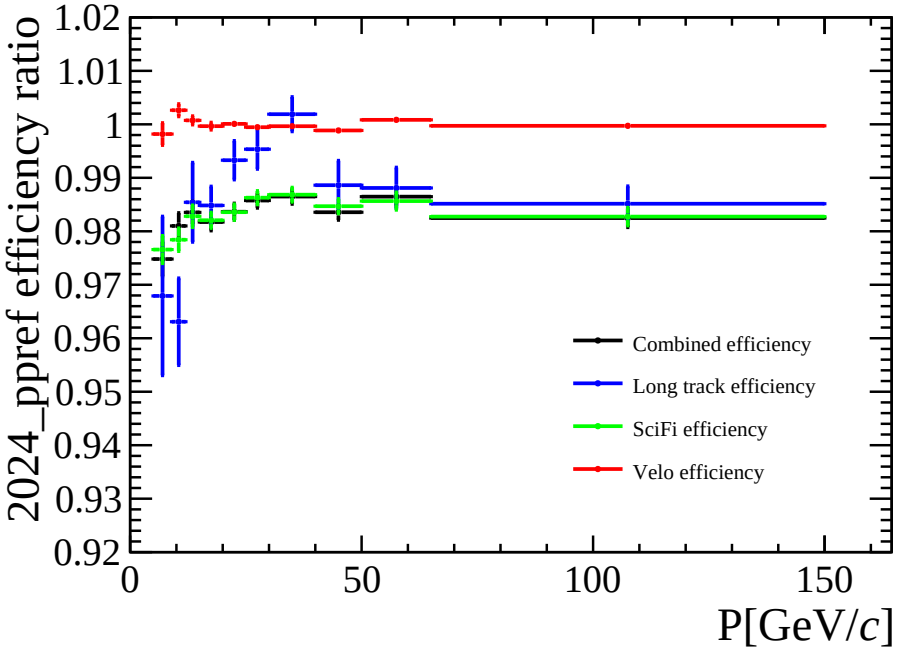


Data/MC ratio for Combined/MuonUT track efficiencies block 8

Tracking efficiency data/MC ratio

Data/MC ratio

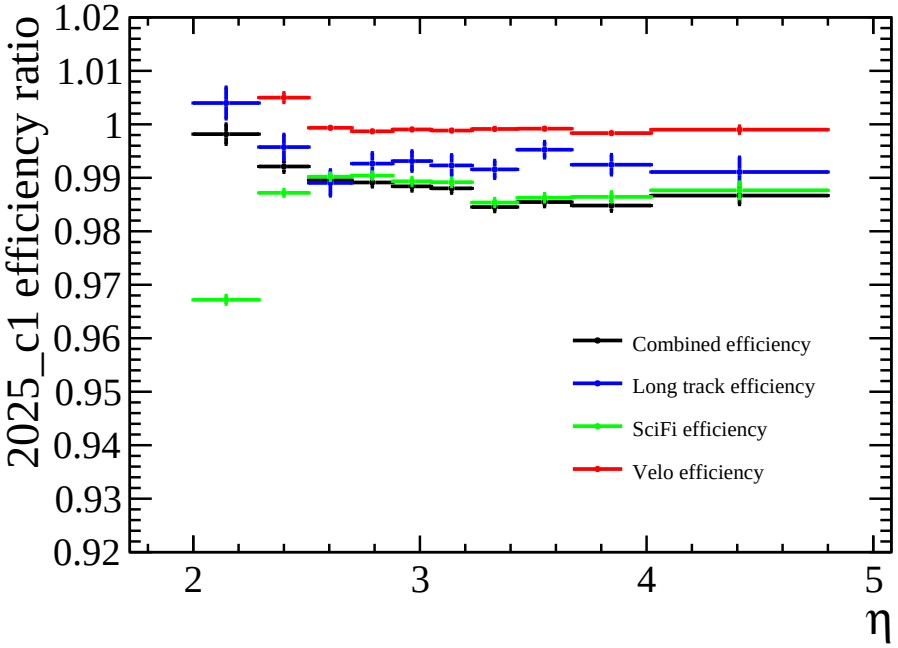


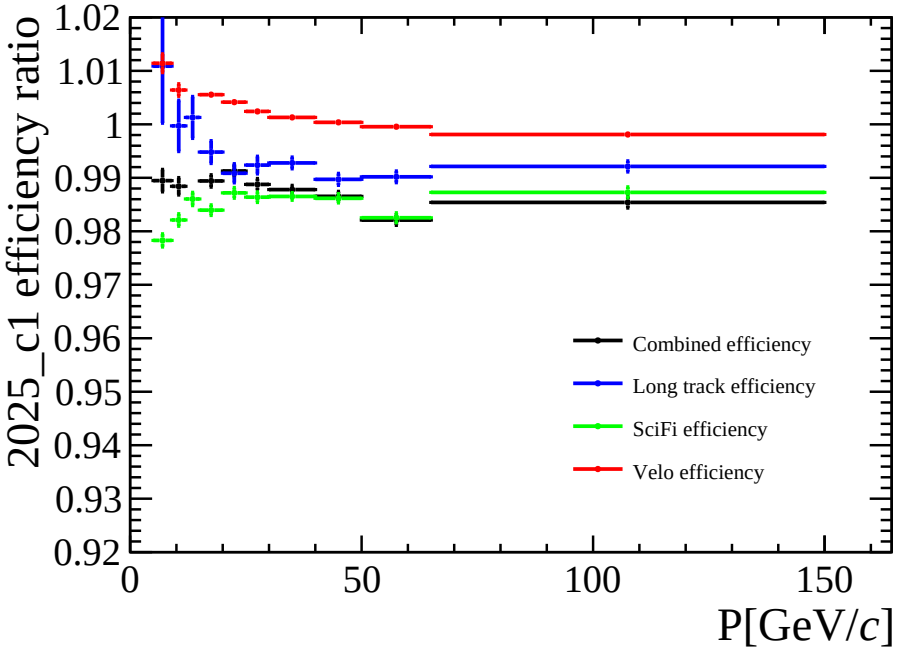


Data/MC ratio for Combined/MuonUT track efficiencies *pp* ref run

Tracking efficiency data/MC ratio

Data/MC ratio





Data/MC ratio for Combined/MuonUT track efficiencies 2025 pre-TS

General observations

General observations on tracking efficiency ratios:

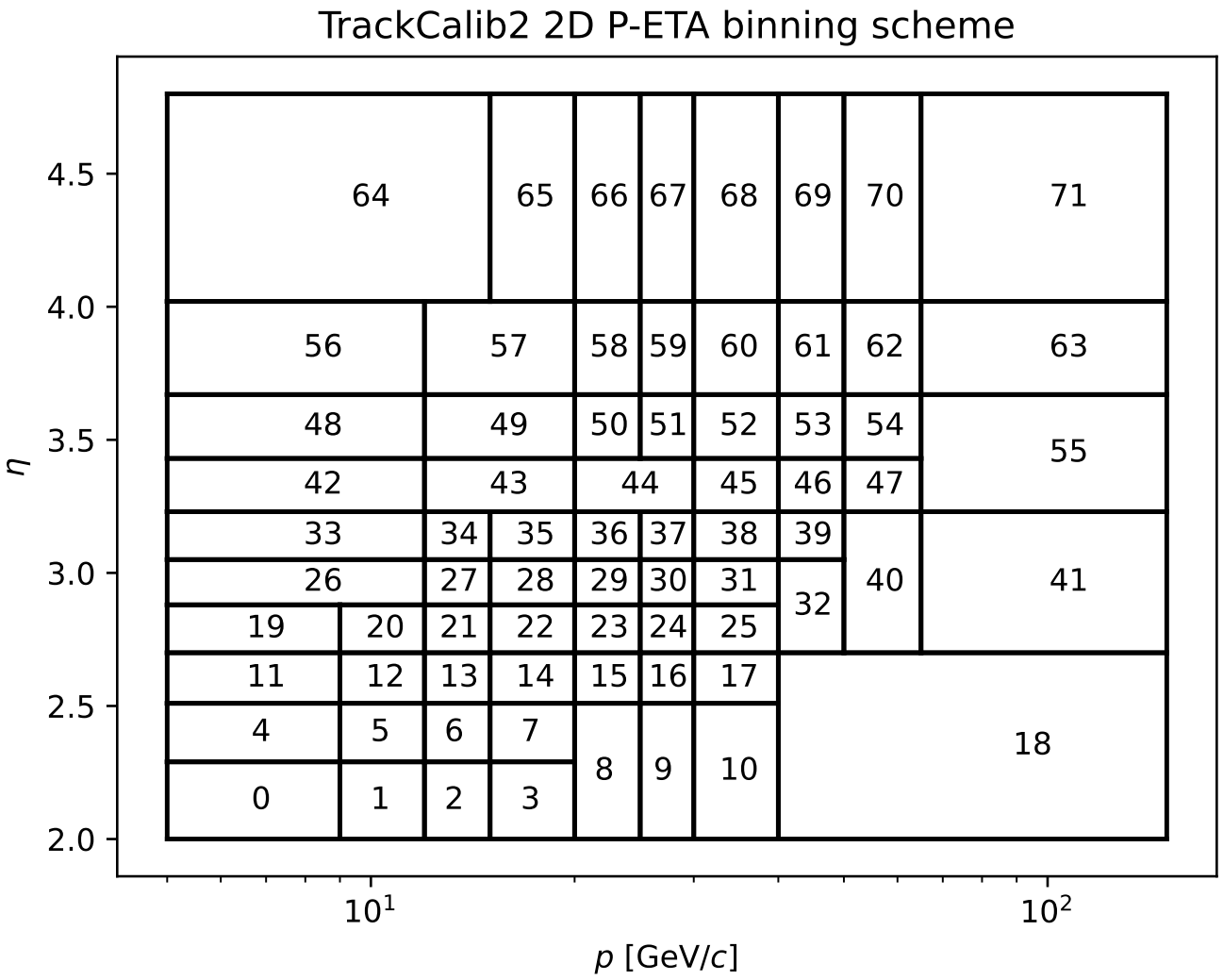
1. Velo efficiencies in 2024 are well described, but underestimated in MC for 2025
2. Combined efficiency is mostly driven by SciFi efficiency
3. Generally there is a 1–2% discrepancy between Combined and MuonUT efficiencies, but effect is larger at low p and low η, so discrepancies are smaller in 2D (next slide)
4. Data–MC differ by almost 5% at low p and low η and needs to be understood

2D binning

- Currently we have 1D bins in p and η with 10 bins each
- Using 10×10 2D bins is impossible
  - Insufficient statistics at high (low) p and low (high) η
  - But most bins <u>do</u> have sufficient statistics
- Strategy: Start with 10×10 bins and merge bins
  - Only consider MC, which is important for constraining the fit shape
  - Only consider VeloMuon yields, as Downstream efficiencies are close to 100% anyway and MuonUT is only used as a cross check
  - Try to merge bins along η at high p and vice versa
  - Aim for roughly $\gtrsim 100$ matched candidates in each bin
- The next plots contain many (tiny) numbers, apologies for this!

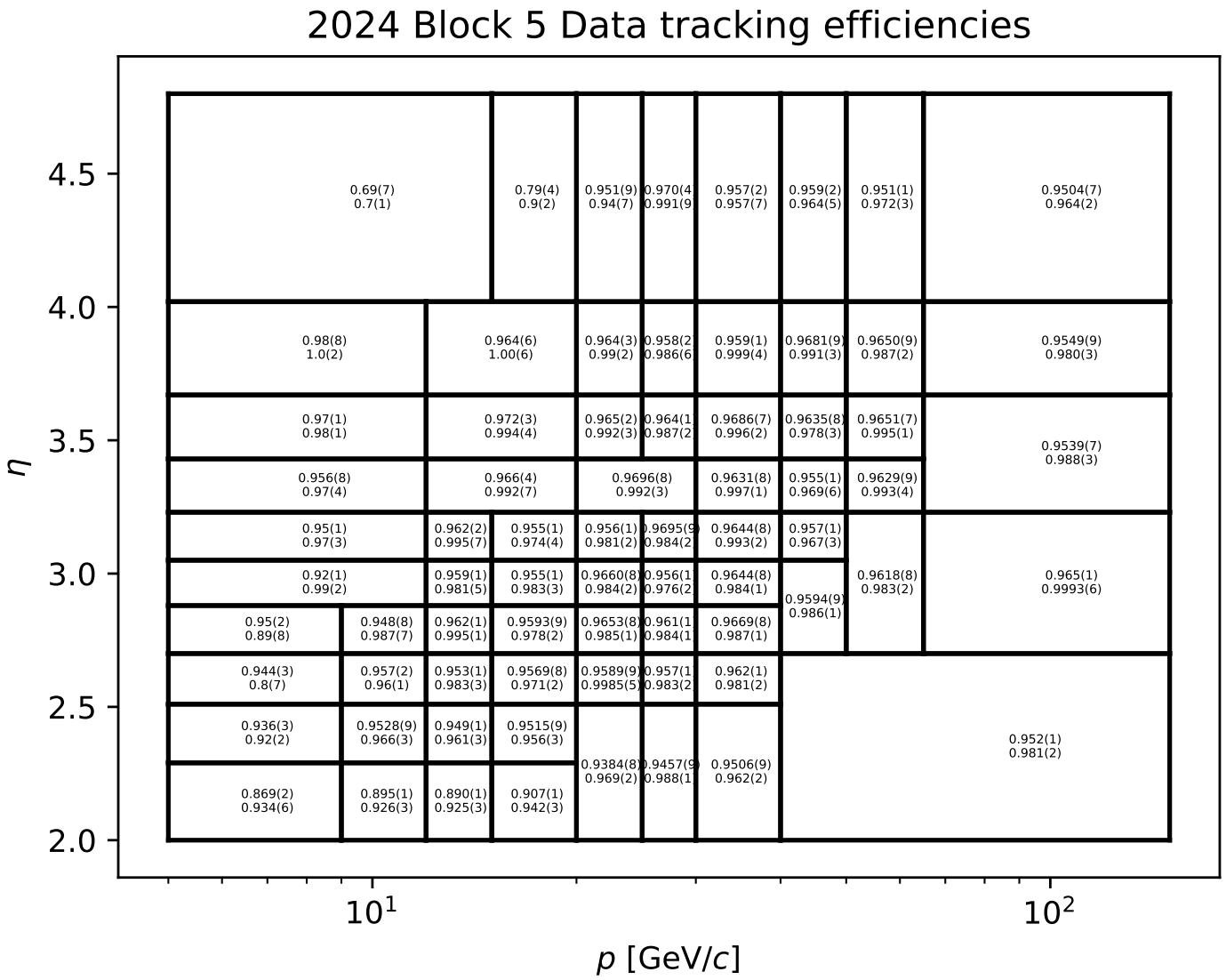
2D binning

Final 2D binning in p vs η with 72 bins:



2D binning

Combined result on top, long track result at the bottom



2D binning

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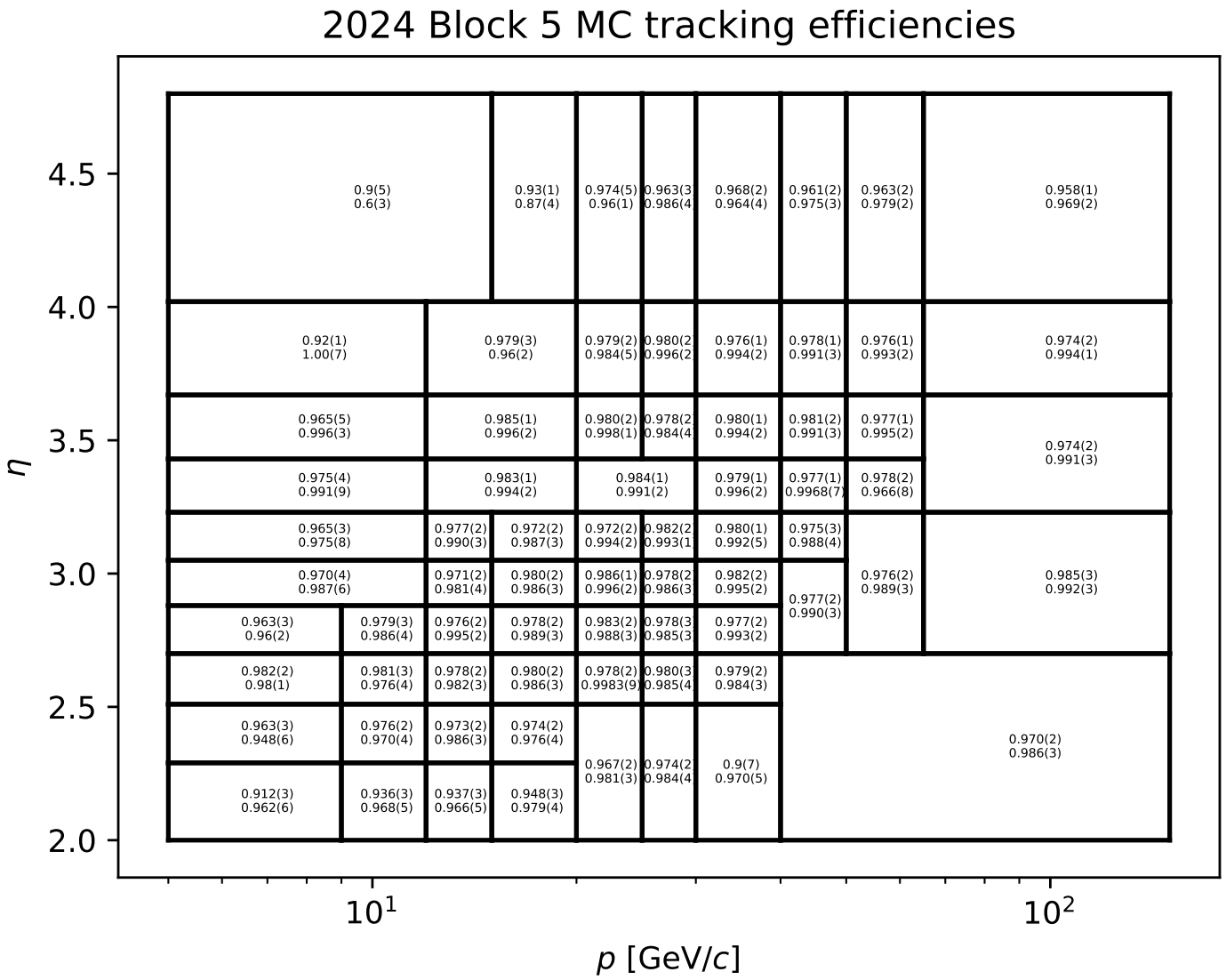

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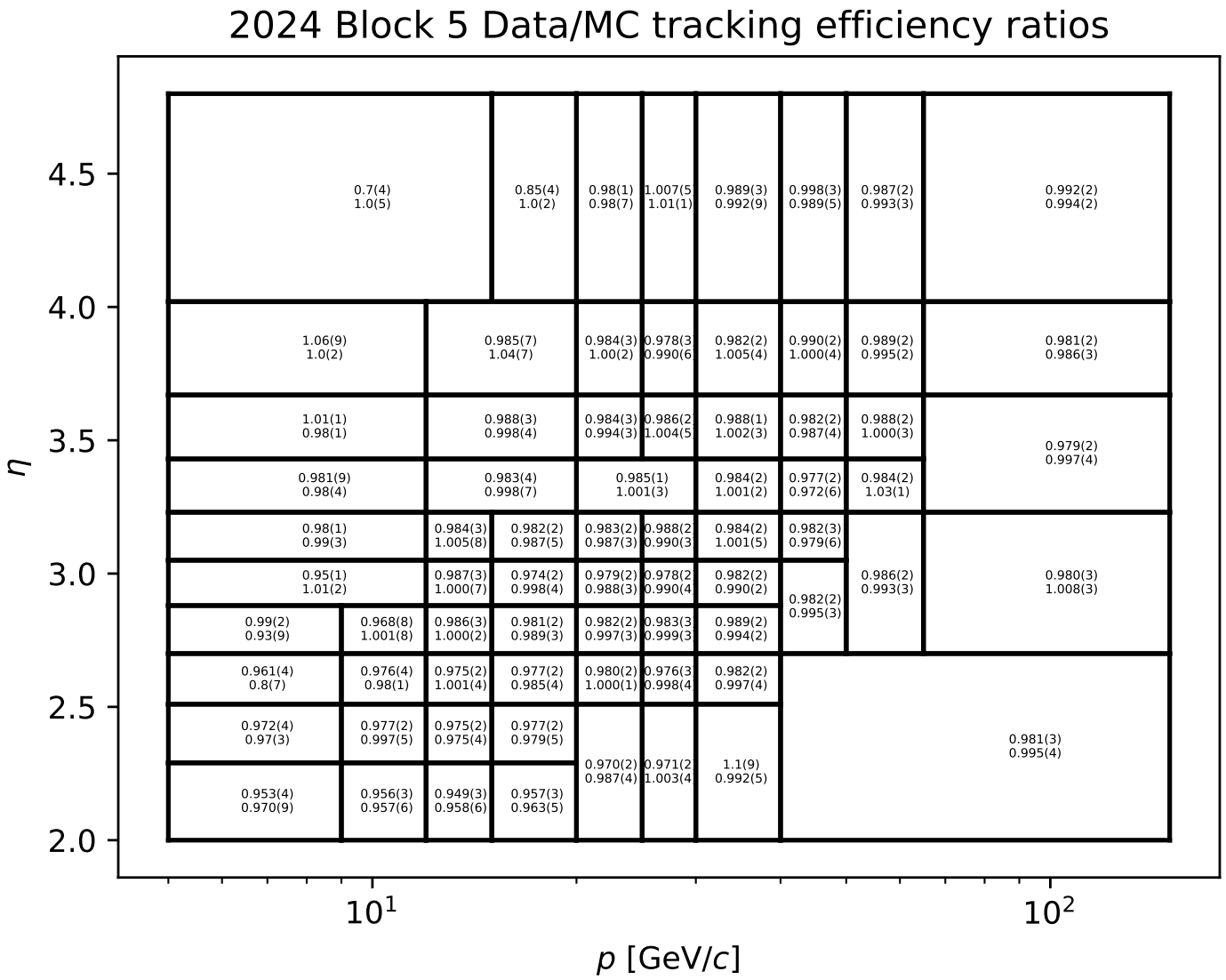
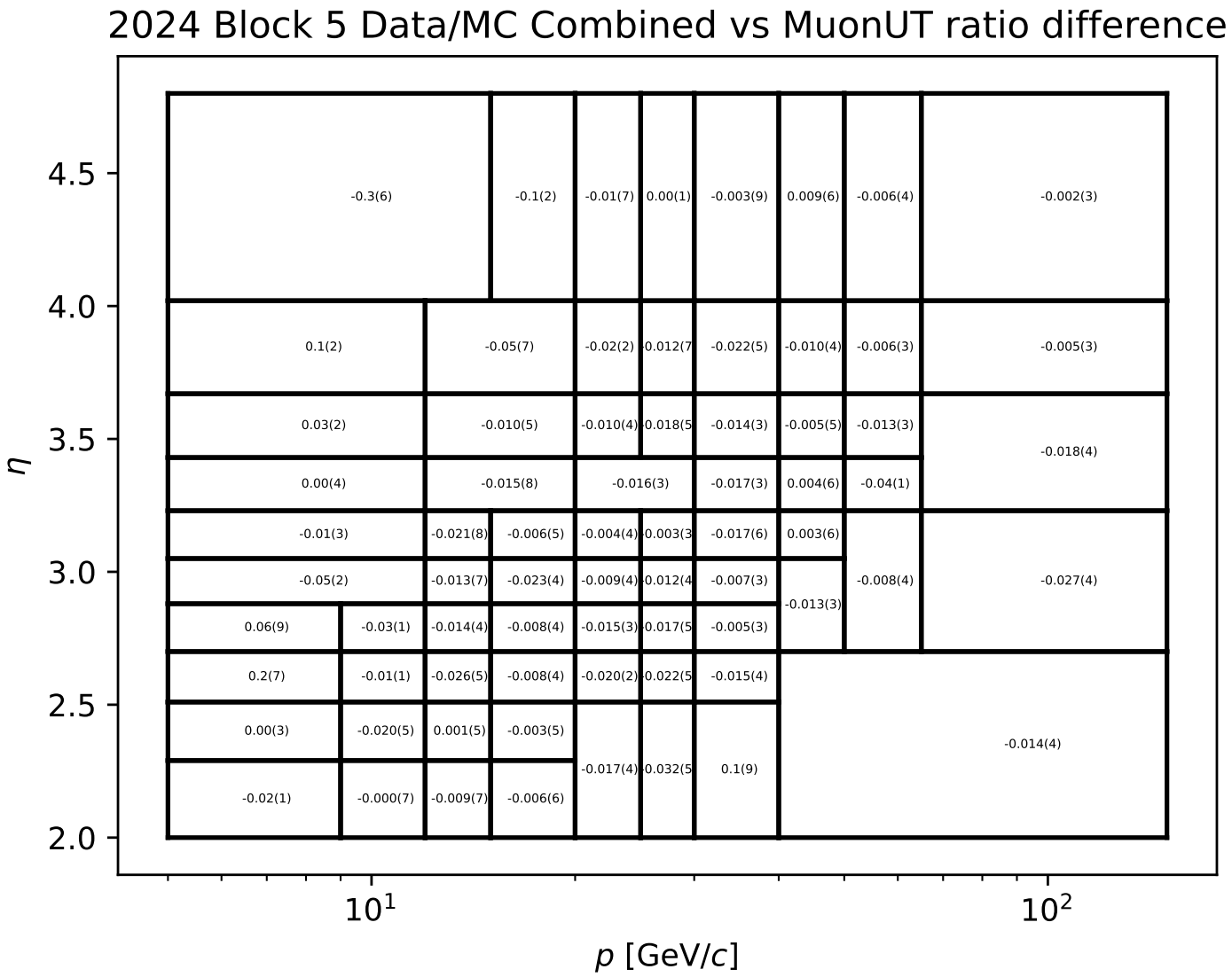


2D binning


2024 Block 5 Data/MC Combined vs MuonUT ratio difference
-0.3(6)
-0.1(2)
-0.01(7)
0.00(1)
-0.003(9)
0.009(6)
-0.006(4)
-0.002(3)
0.1(2)
-0.05(7)
-0.02(2)
0.012(7
-0.022(5)
-0.010(4)
-0.006(3)
-0.005(3)
0.03(2)
-0.010(5)
-0.010(4)
0.018(5
-0.014(3)
-0.005(5)
-0.013(3)
-0.018(4)
0.00(4)
-0.015(8)
-0.016(3)
-0.017(3)
0.004(6)
-0.04(1)
-0.01(3)
-0.021(8)
-0.006(5)
-0.004(4)
0.003(3
-0.017(6)
0.003(6)
-0.05(2)
-0.013(7)
-0.023(4)
-0.009(4)
0.012(4
-0.007(3)
-0.008(4)
-0.027(4)
-0.013(3)
0.06(9)
-0.03(1)
-0.014(4)
-0.008(4)
-0.015(3)
0.017(5
-0.005(3)
0.2(7)
-0.01(1)
-0.026(5)
-0.008(4)
-0.020(2)
0.022(5
-0.015(4)
0.00(3)
-0.020(5)
0.001(5)
-0.003(5)
-0.017(4)
0.032(5
0.1(9)
-0.014(4)
-0.02(1)
-0.000(7)
-0.009(7)
-0.006(6)
4.5
4.0
3.5
3.0
2.5
2.0
η
10^1
10^2
p [GeV/c]

Combined vs MuonUT systematic

Combined vs MuonUT systematic strategy?

- Clearly depends on kinematics, so we should use 2D binning
- It's a method systematic, should not depend on dataset with same μ
- Our proposed strategy:
  1. Determine ratio difference for all data sets in each bin
  2. Combine blocks 5/6 and 7/8
  3. Use blocks 5/6 as proxy for blocks 1/2 due to bug in MuonUT trigger line (see !4047 and !4050)
  4. For bins with a discrepancy above $1\,\sigma$, assign the difference from zero as a systematic
  5. Systematic will most likely be small in most bins, but a small number of bins might have a 1%–2% systematic, in rare cases $> 3\%$

Release of tracking efficiencies

We would like to do an initial release of 2024 tracking efficiencies

- Binning schemes are defined in JSON files
- Data/MC ratio in 2D bins provided for physics analysis, also as JSON
- I'll also make 1D plots in p and η
- Initially release charge integrated numbers, but separate μ^+ and μ^- results will follow
- Where would the most convenient location to put these files?

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2. Results for tracking efficiencies
   - 2024 blocks 2, 1a, 5–8
   - 2024 *pp* ref run
   - 2025 pre-TS
3. **Guide to TrackCalib2**

Guide to TrackCalib2

If you need to produce tracking efficiencies yourselves, for instance if you want to bin in a different variable, you can run TrackCalib2 yourself:

```
git clone ssh://git@gitlab.cern.ch:7999/lhcb-rta/trackcalib2.git
git checkout mtat/dev/2024
git submodule init
git submodule update
lb-conda trackcalib2
python trackcalib.py prepare -mode MC -year 2024_Block5 -method VeloMuon
python trackcalib.py fit -mode MC -year 2024_Block5 -method VeloMuon -variables "P-ETA"
```

Summary and next steps

1. Block 1b MC doesn't describe SciFi FE exclusion well
2. Ready for initial release of 2024 blocks 2, 1a, 5–8, *pp* ref
   - Several issues have been resolved
   - Behaviour consistent between 2024 blocks
3. 2025 Velo efficiencies in MC need to be understood
4. Additional work required for next release:
   - Larger MC samples (maybe 4M $\rightarrow$ 10M), requires PPG permission
   - Include `nFTClusters` reweighting using work by Maurice
   - Other systematics